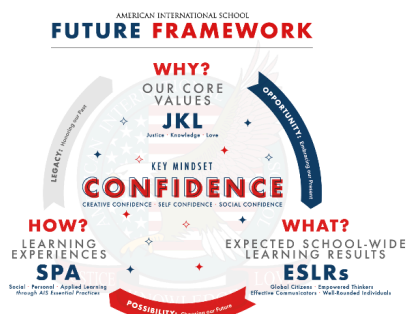




KEY MINDSET CONFIDENCE

LEGACY · Honoring our Past
OPPORTUNITY · Embracing our Present
POSSIBILITY · Choosing our Future

KEY TEACHING & LEARNING EXPERIENCES AT AIS SOCIAL · PERSONAL · APPLIED

SCHOOL MISSION:
To provide a comprehensive and challenging academic program that develops life-long learners and prepares students to meet the responsibilities and expectations of global citizenship.

OUR PURPOSE:
To create students who are Global Citizens, Empowered Thinkers, Effective Communicators, and Well-Rounded Individuals.

AIS WHOLE SCHOOL NORMS

- AIS Core Values drive decisions and actions
- Ensure all voices are heard
- Assume best intentions
- Always go to the source
- Start and end on time
- Be fully present
- Tough on ideas and soft on people
- We are a caring community

OUR CORE VALUES: **Justice** **Knowledge** **Love**

GEOMETRY

Unit 7: Solid Geometry

Final Assessment

Full Name:	Grade:	Date:
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Learning Standards:

HS.M.AG.C.1 - Use clear and logical justification when solving problems and explaining your understanding.

HS.M.G.1.6.G.A.4 - 6.G.A.4 Represent three-dimensional figures using nets made up of rectangles and triangles. [1]

HS.M.G.5.G.MG.A.1 - G.MG.A.1 Use geometric shapes, their measures, and their properties to describe objects. [1]

HS.M.G.5.G.STR.D.9 - G.STR.D.9 Derive the formula $A = \frac{1}{2} ab \sin(C)$ for the area of a triangle by drawing an auxiliary line from a vertex perpendicular to the opposite side. [2,3]

HS.M.G.5.G.GPE.B.7 - G.GPE.B.7 Use coordinates to compute perimeters of polygons and areas of triangles and rectangles, e.g., using the distance formula. [4,5]

HS.M.G.5.G.GMD.B.4 - G.GMD.B.4 Identify the shapes of two-dimensional cross-sections of three-dimensional objects, and identify three-dimensional objects generated by rotations of two-dimensional objects. [6,7]

HS.M.G.5.G.GMD.A.1 - G.GMD.A.1 G.GMD.A.1 Give an informal argument for the formulas for the circumference of a circle, area of a circle, volume of a cylinder, pyramid, and cone. [8,9]

HS.M.G.5.G.GMD.A.3 - G.GMD.A.3 Use volume formulas for cylinders, pyramids, cones, and spheres to solve problems. [10,11]

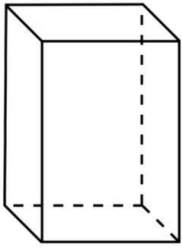
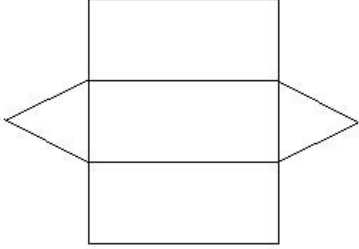
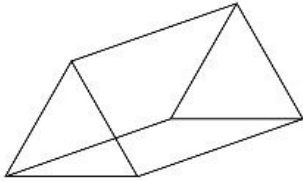
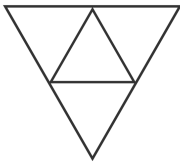
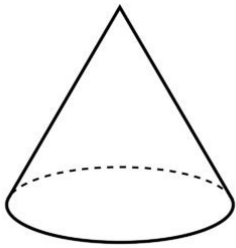
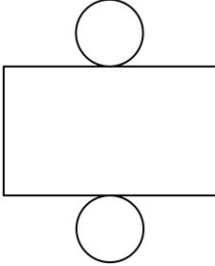
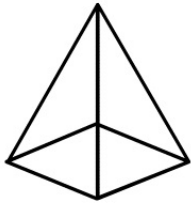
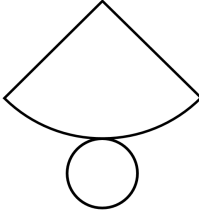
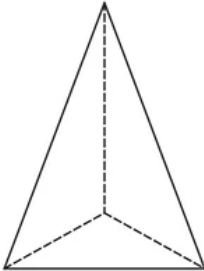
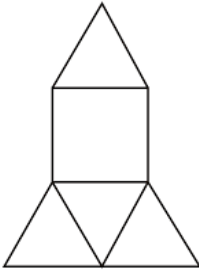
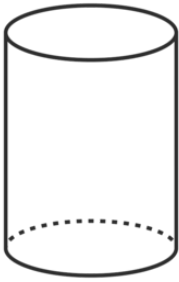
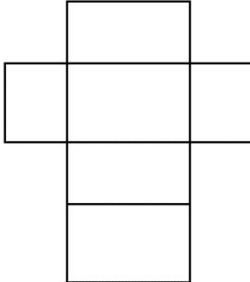
HS.M.G.5.G.MG.A.2 - G.MG.A.2 Apply concepts of density based on area and volume in modeling situations (e.g., persons per square mile, BTUs per cubic foot). [12,13]

INSUFFICIENT EVIDENCE	BEGINNING PROFICIENCY	APPROACHING PROFICIENCY	DEMONSTRATING PROFICIENCY	MASTERING PROFICIENCY
I am unable to solve basic problems. I can never apply this standard when solving problems. I am inaccurate when explaining my thinking.	I am beginning to solve basic problems. I can rarely apply this standard when solving problems. I am minimally accurate when explaining my thinking.	I can solve rudimentary problems including those involving simple scenarios. I can sometimes apply this standard when solving problems. I can somewhat accurately explain my thinking.	I can solve familiar problems including those involving real-world scenarios. I can usually apply this standard when solving problems. I am generally accurate when explaining my thinking.	I can solve unfamiliar problems, and analyze unique solutions. I can consistently apply this standard when solving problems. I can accurately and coherently explain my thinking. I can make connections between this concept and other areas of mathematics.

Direction:

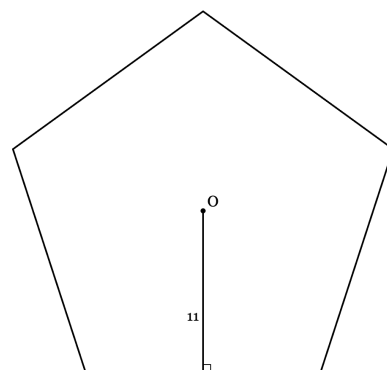
- Solve each of the following problems.
- Do not spend too much time on one problem.
- Write all the steps to arrive at the final answer.

01 Match the following solid with its corresponding net by writing (a) to (f) and write the solids name.

Solid	Net	Name
		
		
		
		
		
		

02

O is the center of the regular pentagon below. Find its area. Round your answer to the nearest tenth. Show all the work required to arrive at your final answer.

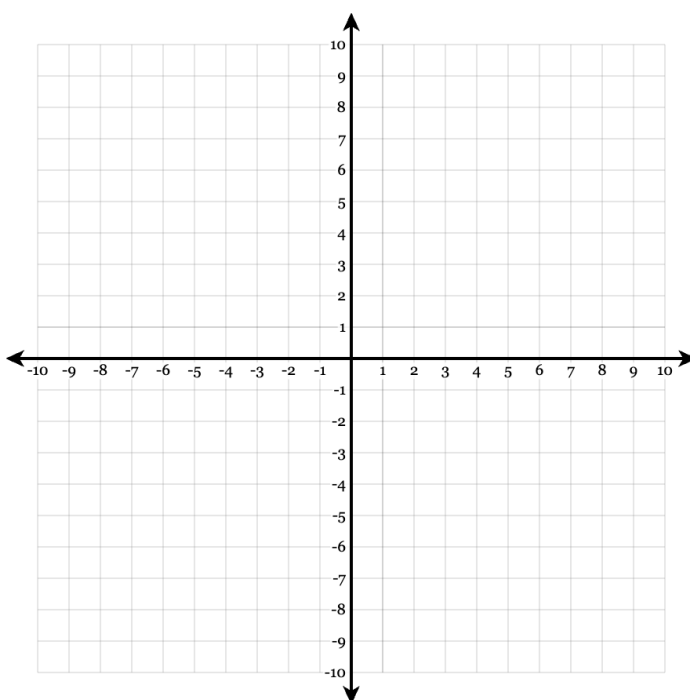


03

Two sides of a parallelogram are 28 feet and 84 feet . The measure of the angle between these sides is 66° . Find the area of the parallelogram to the nearest square foot. Show all the work required to arrive at your final answer.

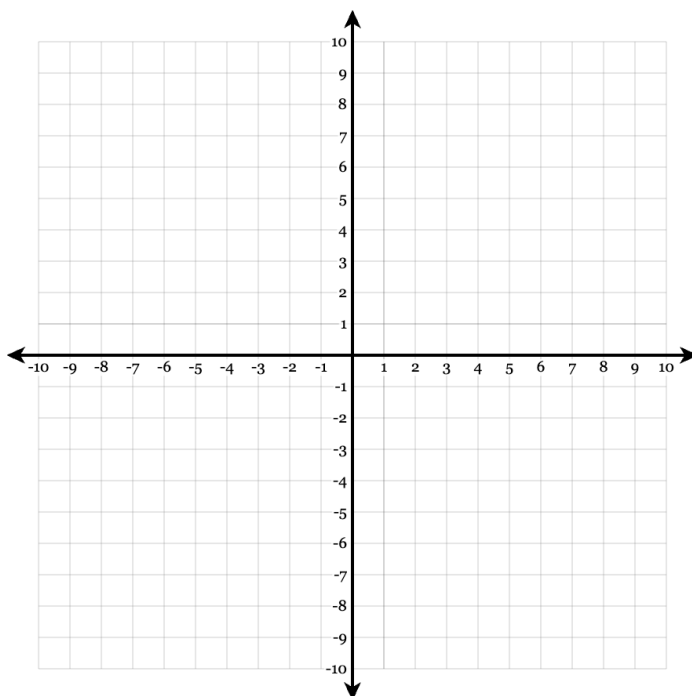
04

The points $O(6, -4)$, $P(6, 2)$, $Q(1, 7)$, and $R(1, 1)$ form parallelogram $OPQR$. Plot the points. Find the area of the parallelogram. Show all the work required to arrive at your final answer.



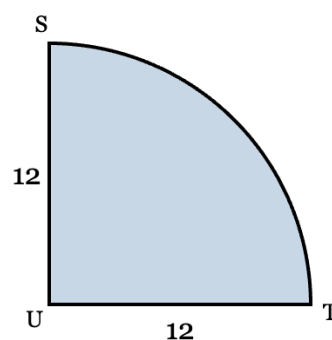
05

The points $M(-3, -4)$, $N(4, 0)$, $O(5, 8)$, and $P(-2, 4)$ form rhombus $MNOP$. Plot the points. Find the area of the rhombus. Show all the work required to arrive at your final answer.



06

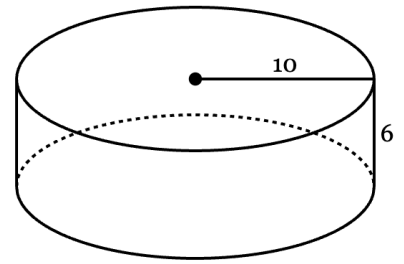
The figure below is a sector of a circle. Name the solid generated when the sector is rotated about $\overline{UT}$. Calculate the volume. Round your answer to the nearest tenth. Show all the work required to arrive at your final answer.



07

What type of cross-section is formed when the right hexagonal prism is cut (i) parallel (ii) perpendicular to its base?

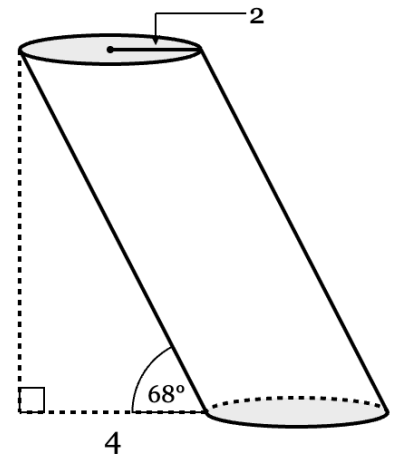
- 08 The solid below is dilated by a scale factor of $\frac{1}{4}$. Find the surface area of the solid created upon dilation. Answer in terms of π . Show all the work required to arrive at your final answer.



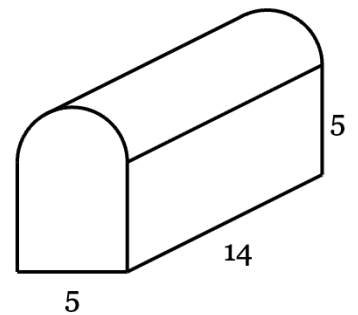
- 09 A company is designing boxes in the shape of rectangular prisms. The company makes a small box that contains 1.2 ounces. Note: Ounces are a measure of volume.

- (a) The extra-large box will be a dilation of the small box using a scale factor of 4. How many ounces will the extra-large box contain? Explain or show your reasoning.
- (b) Another box will be a dilation of the small 1.2 — ounce box. It must contain 18.75 ounces. What scale factor does the company need to use for this box? Explain or show your reasoning.
- (c) The small box's volume of 1.2 ounces is about 2 cubic inches. The large box will be a dilation of the small box using a scale factor of 3. Find a possible set of dimensions for the large box. Round your answer to the nearest tenth if necessary. Explain or show your reasoning.

- 10 Find the volume of the oblique cylinder below. Round your answer to the nearest tenth.



- 11 A loaf of bread can be modeled by a rectangular prism topped with half a right cylinder. Sophia measures the dimensions of the prism as 5in by 5in by 14in . Using those dimensions, find the volume of the loaf. Round your answer to the nearest tenth. Show all the work required to arrive at your final answer. (Note: diagram is not drawn to scale.)



- 12 Which of the states in the table below has the least area? Show all the work required to arrive at your final answer.

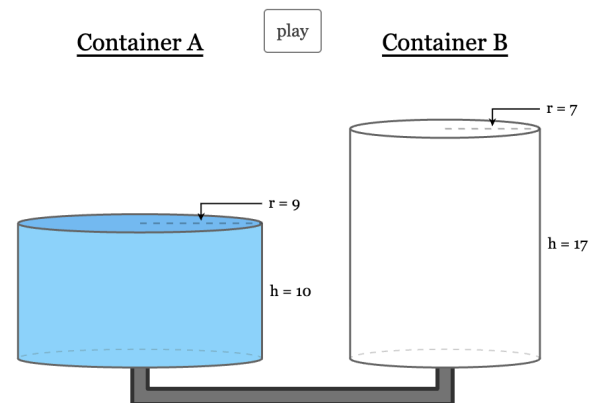
State	Population in 2020	Population Density (people/mi ²)
Arkansas	3,011,524	56.63
South Dakota	886,667	11.50
Utah	3,271,616	38.54
New Mexico	2,117,522	17.42

13

A container built for transatlantic shipping is constructed in the shape of a right rectangular prism. Its dimensions are 7 ft by 8.5 ft by 10 ft . The container is entirely full. If, on average, its contents weigh $0.85\text{ pounds per cubic foot}$, and on average, the contents are worth $\$3.04\text{ per pound}$, find the value of the container's contents. Round your answer to the nearest cent. Show all the work required to arrive at your final answer.

14

Two containers designed to hold water are side by side, both in the shape of a cylinder. Container A has a radius of 9 feet and a height of 10 feet. Container B has a radius of 7 feet and a height of 17 feet. Container A is full of water and the water is pumped into Container B until Container A is empty. After the pumping is complete, what is the percent Container B that is empty, to the nearest tenth? Show all your work that leads you to the final answer.



END OF THE ASSESSMENT